

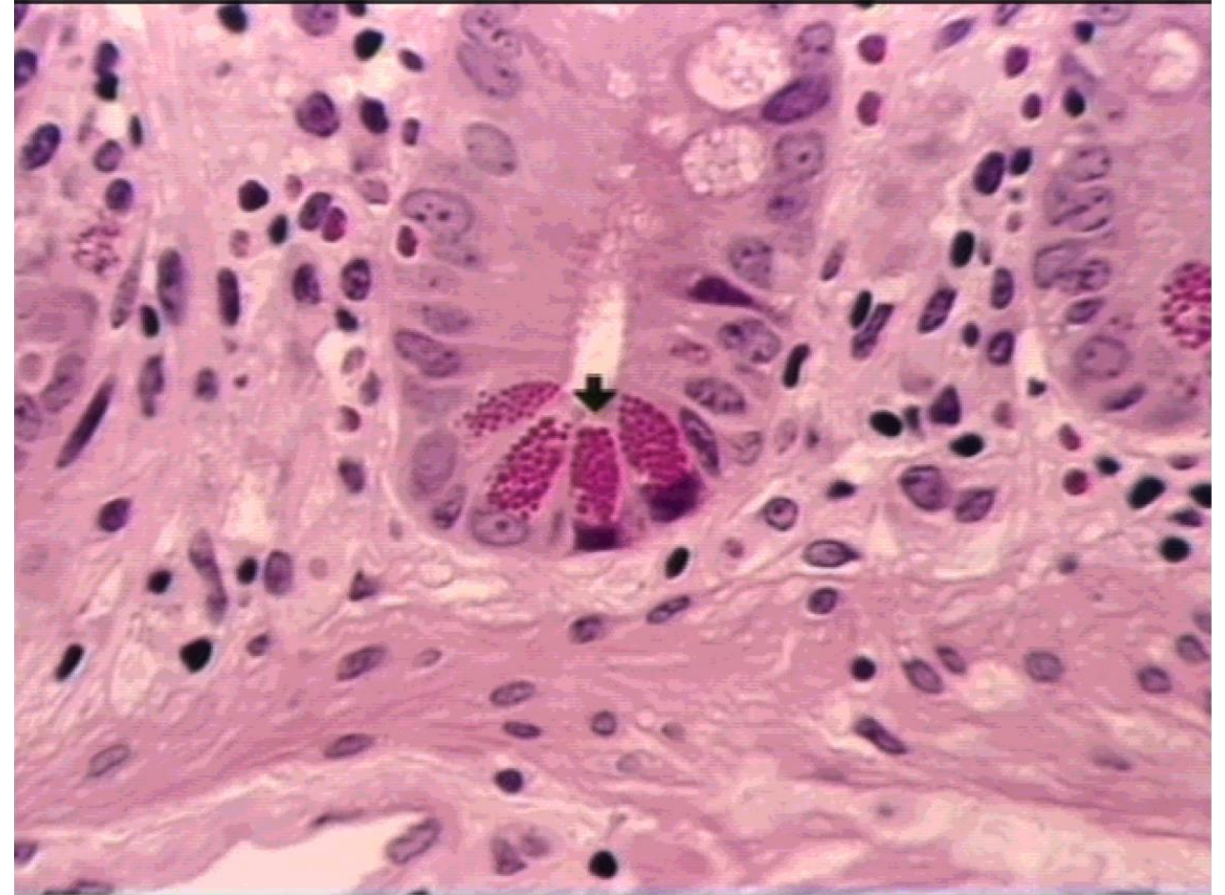


DIGESTION AND METABOLISM

IMCQ 4 ANAT & HCB

1. A 38-year-old man comes to the physician for an endoscopy. On visualization of small intestine, some areas of the mucosa are found to be inflamed. Biopsy of this region is taken, and the tissue is sent for histopathological examination. The tissue is found to be normal and shows several glands as shown below. Which of the following functions is most likely performed by the cells at the pointer?

- A. Regulates intestinal normal bacterial flora
- B. Unicellular mucus secreting cells
- C. Produce digestive enzymes
- D. Produce secretin and motilin
- E. Serve as stem cells of the epithelium



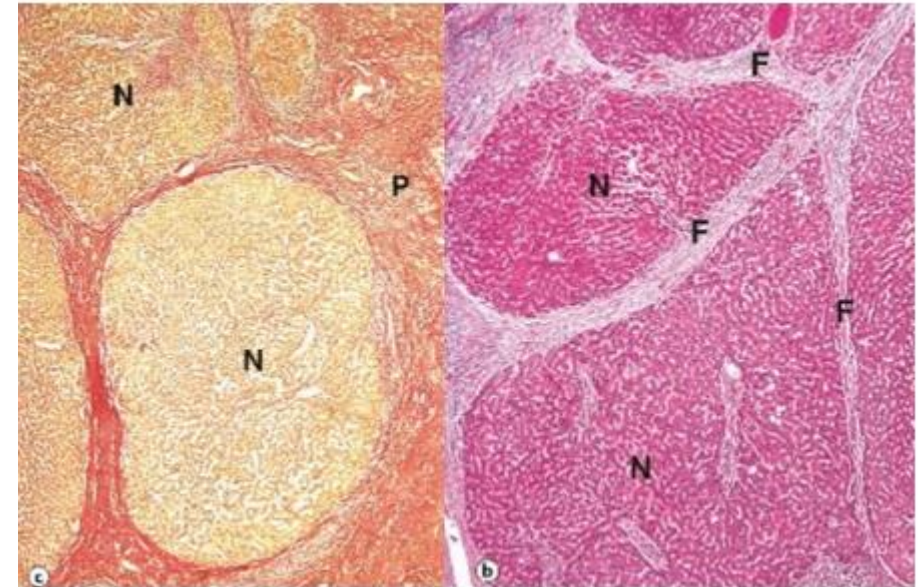
2. A 58-year-old man is brought to the emergency department because of a 5-hour history of severe abdominal pain, nausea and vomiting. Physical examination shows fever, abdominal distension and nonlocalized pain. Laboratory studies show leukocytosis and sepsis. A CT of the abdomen shows a blockage of the superior mesenteric artery and he undergoes surgical resection of the bowel. The segment that is affected has the histological appearance as shown in the attached image. Which of the following best describes the area of the gut affected?

- A. Contains cells that produce intrinsic factor
- B. Contains large number of Paneth cells
- C. Contains glands in the submucosa
- D. Produces vitamin B12
- E. Omental appendices can be found on the antimesenteric border
- F. Contains limited number of goblet cells



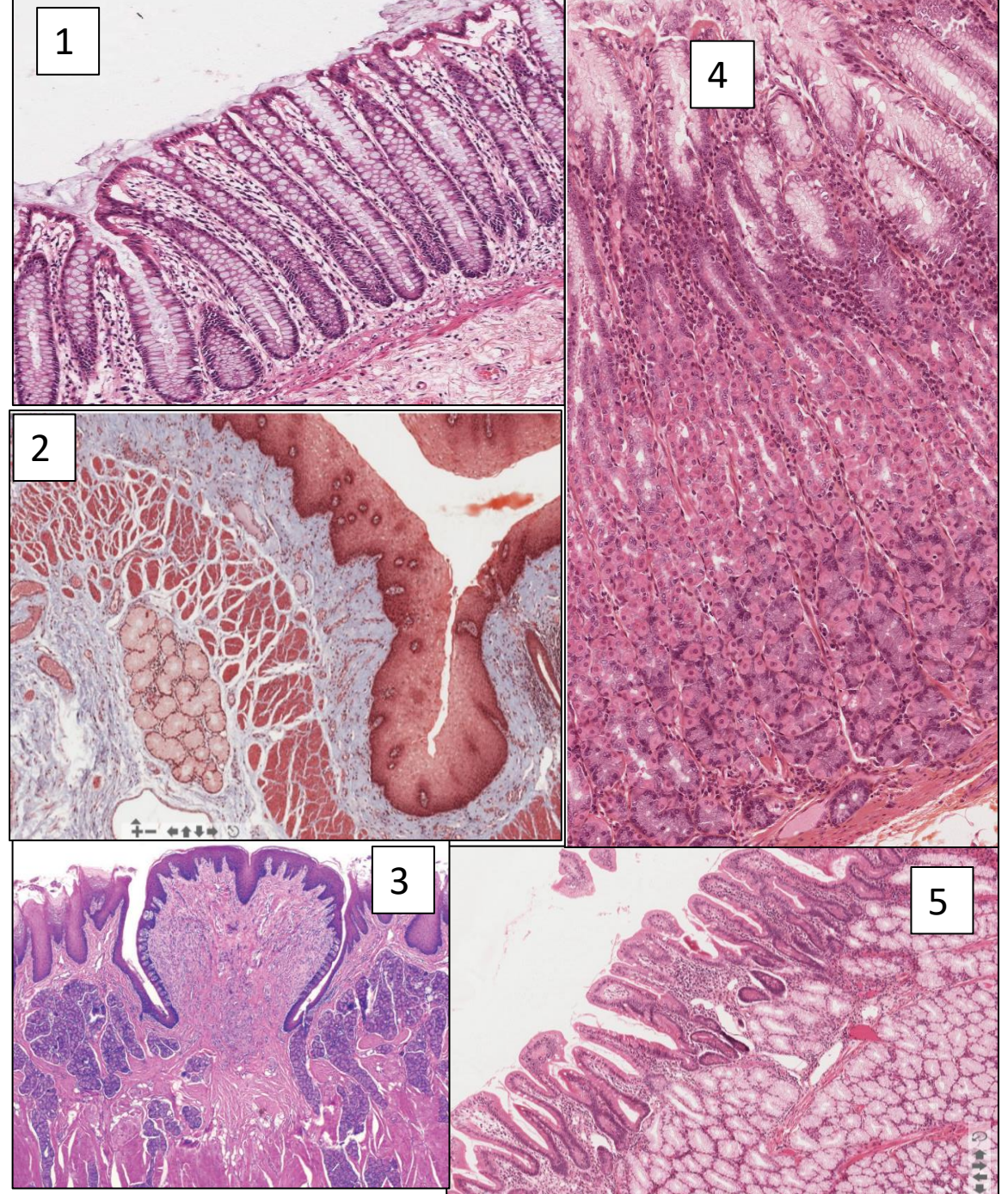
3. A 46-year-old man collapses on the sidewalk and is brought to the emergency department. Physical examination shows that he is febrile, jaundiced and has a firm globally distended abdomen. Laboratory studies show elevated serum bilirubin, low plasma proteins and grossly elevated liver enzymes. He dies despite appropriate medical care. During the autopsy, a section of the liver is prepared for histopathological analysis and the image is shown below. Which of the following cells are most likely responsible for laying down the fibrous tissue seen at “F” in the attached image?

- A. Parietal cell
- B. Kupffer cell
- C. Ito cell
- D. G cells
- E. Hepatocyte
- F. Paneth cell
- G. Cholangiocyte



4. A 34-year-old man comes to the physician because of a 6-month history of weakness and weight loss. Physical examination shows pallor and emaciation. Peripheral blood smear shows a macrocytic anemia consistent with pernicious anemia. Which of the following areas of the GI tract in the labeled image was most likely damaged to result in his condition?

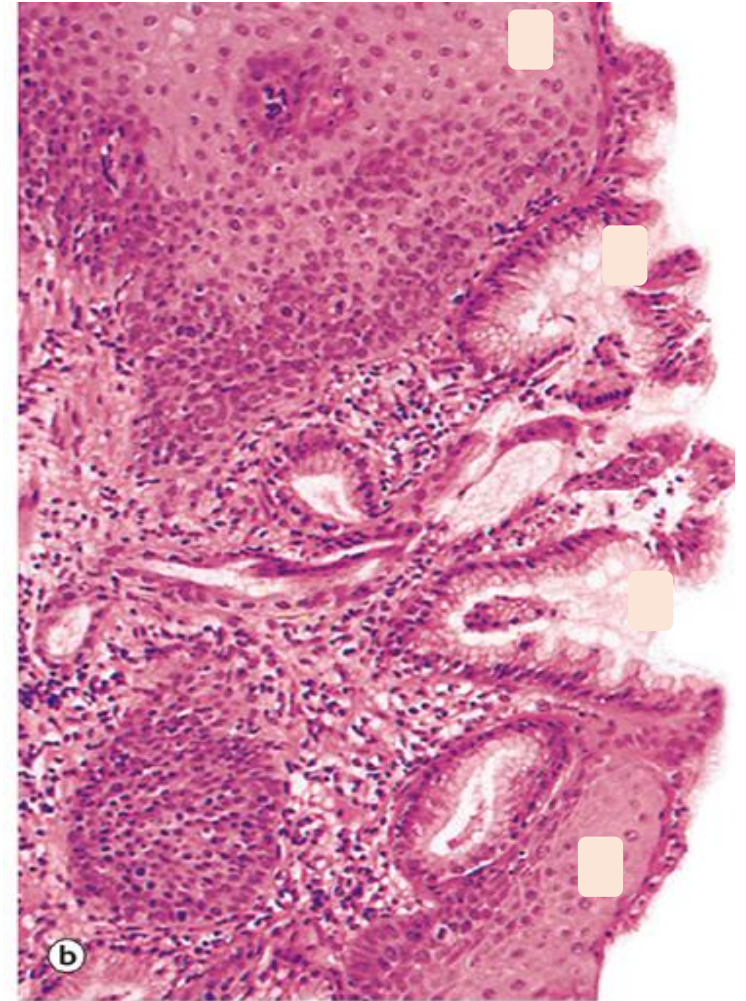
- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



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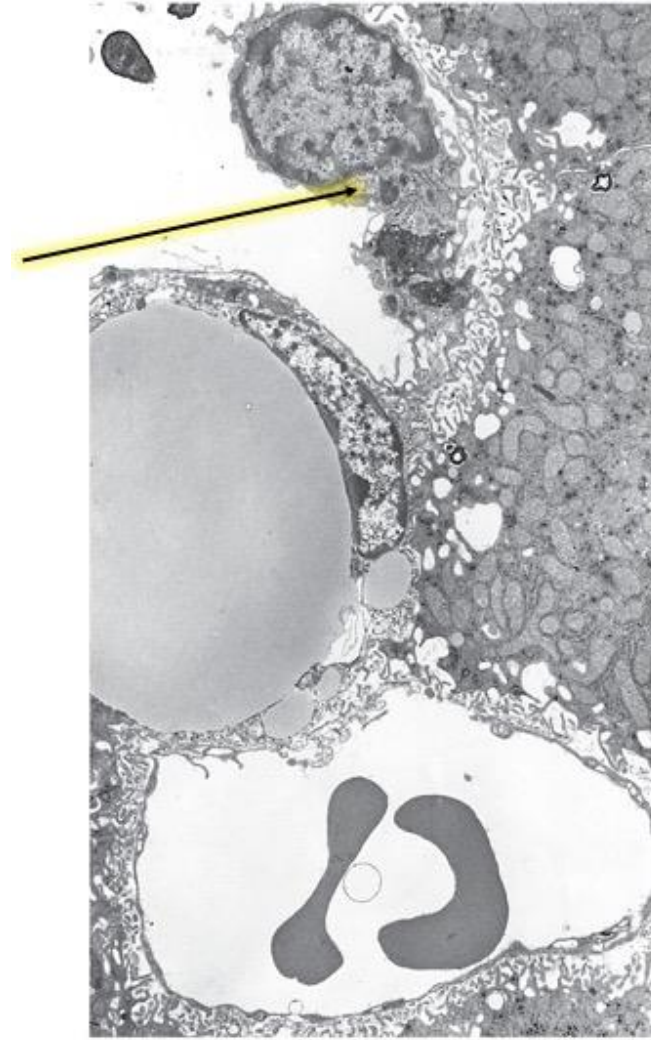
5. A 43-year-old man comes to the emergency department because of a 1-week history of a burning sensation in his chest and difficulty swallowing. He has a 3-year history of heartburn. Physical examination shows no abnormalities. Endoscopy of the upper GI shows inflammation and a biopsy is collected from the lower end of esophagus which is shown. Which of the following cellular adaptations to injury most likely occurred in this patient's condition?

- A. Atrophy
- B. Hyperplasia
- C. Dysplasia
- D. Metaplasia
- E. Hypertrophy



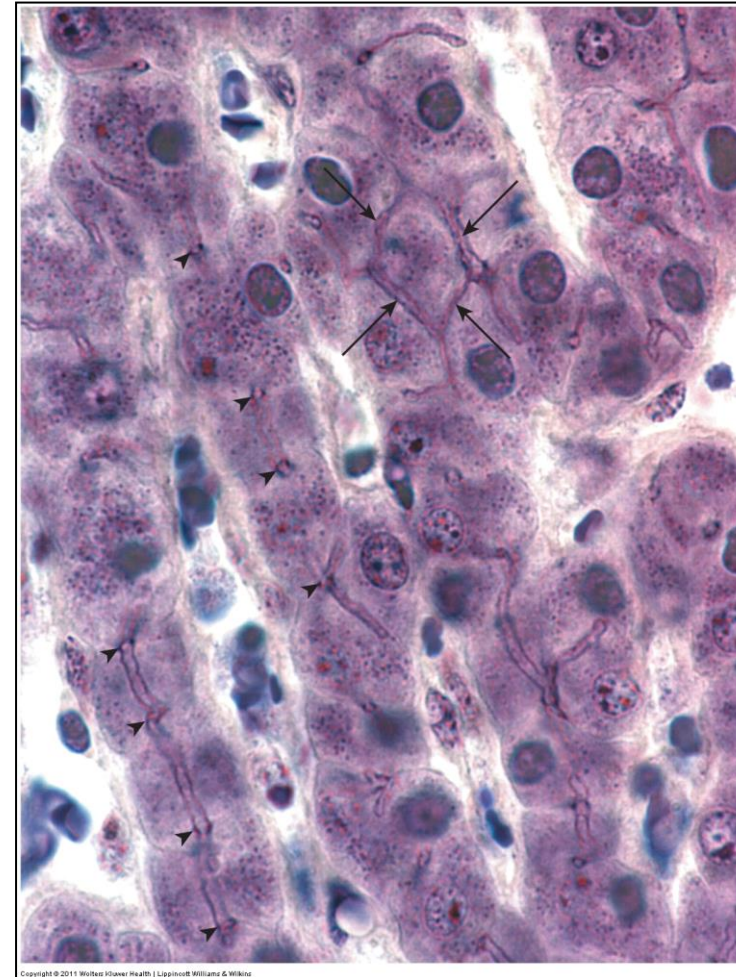
6. A 72-year-old man comes to the physician because of a 2-week history of upper abdominal pain. Physical examination shows enlarged liver. A biopsy is collected from the liver and a TEM of a hepatic sinusoid is attached for reference. Which of the following is cells is most likely shown on the image at the pointer?

- A. Endothelial cell
- B. Kupffer cell
- C. Hepatic stellate (Ito) cell
- D. Hepatocyte
- E. Cholangiocyte



7. A 65-year-old man comes to the physician because of a 1-month history of upper abdominal pain. Physical examination shows enlarged liver and jaundice of mucous membranes. Blood tests show elevated liver enzymes. A liver biopsy shows obstruction of the structures at the arrows & arrowheads. Which of the following cells most likely form these structures?

- A. Endothelial cells
- B. Hepatic stellate (Ito) cells
- C. Kupffer cells
- D. Cholangiocytes
- E. Hepatocytes

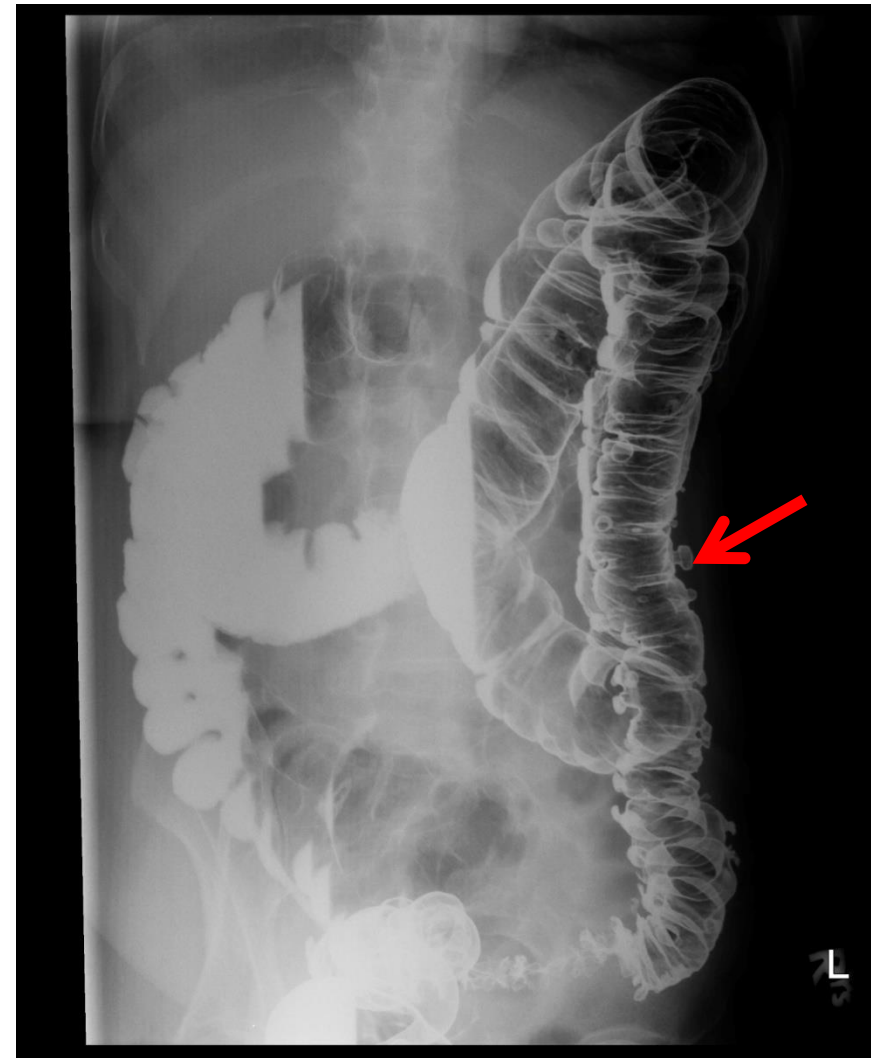


8. A 43-year old woman comes to the physician because of a 2-week history of lower abdominal pain. Physical examination shows a distended, painful abdomen with absent bowel sounds. She has had previous surgeries for gallstones, perforated gastric ulcer and appendicitis. Surgical exploration shows severe intestinal adhesion formation. Which of the following histological layers of the tubular GI tract was most likely damaged during her past surgeries resulting in these adhesions?

- A. Adventitia
- B. Muscularis mucosa
- C. Serosa
- D. Submucosa
- E. Muscularis externa
- F. Lamina propria

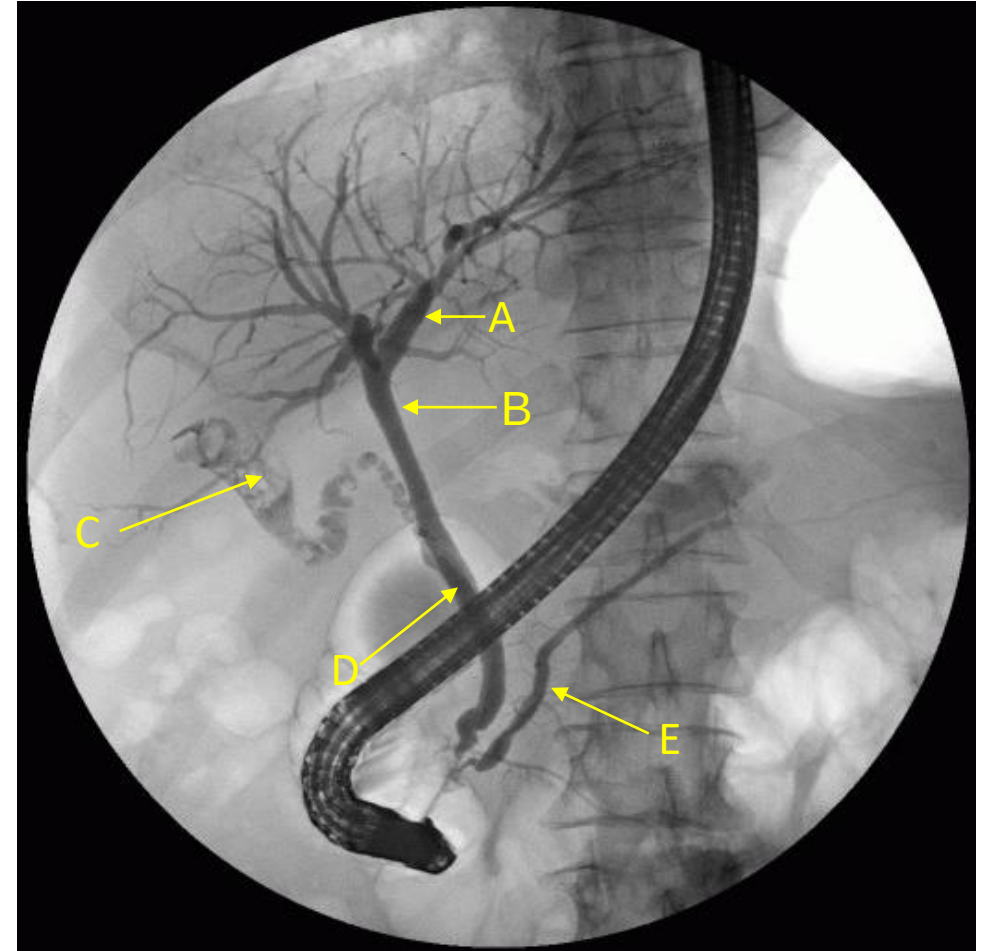
9. A 60-year-old woman comes to the emergency department because of a 2-day history of fever and severe abdominal pain. Physical examination shows muscle guarding and rebound tenderness on deep palpation. An x-ray and barium enema of her abdomen is shown. Which of the following nerves most likely carries the visceral afferent fibers from the part of the GI indicated by the arrow back to the spinal cord?

- A. Lumbar splanchnics
- B. Pelvic splanchnics
- C. Lesser thoracic splanchnics
- D. Greater thoracic splanchnics
- E. Least thoracic splanchnics
- F. Vagus



10. A 40-year-old woman comes to the emergency department because of a 12-hour history of nausea, vomiting, intense itching and pain in the right upper quadrant. Physical examination shows yellow mucous membranes, abdominal tenderness and inspiratory arrest on deep palpation in the right subcostal region. Laboratory studies show elevation of serum bilirubin and markedly increased alkaline phosphatase. Endoscopic retrograde cholangiopancreatography(ERCP) shows an obstruction within the biliary tree. In the image provided, which of the following locations most likely represents the site of obstruction?

- A. A
- B. B
- C. C
- D. D
- E. E



11. A 72-year-old man is brought to the emergency department because of a 1-hour history of severe abdominal pain. His blood pressure is 70/60mmHg, respiratory rate is 22/min and pulse is 100/min. Physical examination shows a pulsatile mass on superficial palpation of the abdomen. CT scan shows a large abdominal aortic aneurysm that includes the origin of the inferior mesenteric artery. During surgical repair of the aneurysm, the inferior mesenteric artery is sacrificed. Which of the following arteries would most likely provide collateral supply to the descending colon?

- A. Left gastroepiploic
- B. Right colic
- C. Splenic
- D. Superior rectal
- E. Inferior rectal

12. A 55-year-old man comes to the physician because of a 6-month history of gradually increasing fatigue and edema despite diuretic therapy. His supine blood pressure is 95/48 mmHg. Physical examination shows dependent edema to the mid-thigh and fine crepitations at the base of the lungs. Laboratory studies show impaired renal function. Imaging studies shows globally decreased blood flow including to the GI arteries resulting in generalized ischemia. Which part of the colon will most likely show the greatest amounts of ischemic changes?

- A. Cecum
- B. Ascending colon
- C. Right colic flexure
- D. Left colic flexure
- E. Transverse colon

13. A 60-year-old woman is brought to the emergency department because of a 1-hour history of hematemesis. She has had multiple previous admissions due to chronic alcoholism. Physical examination shows pallor, yellow sclerae, abdominal distention and dilated veins on the anterior abdominal wall. Endoscopy shows bleeding esophageal varices. Attempts to ligate the varices are unsuccessful and follow up surgery is planned. Which of the following venous anastomoses will most likely reduce the pressure in the portal system thereby stopping the bleeding?

- A. Superior mesenteric to inferior mesenteric
- B. Left gastric to splenic
- C. Right renal to right gonadal
- D. Portal to IVC
- E. Right gastric to right gastro-omental

14. A 2-year-old girl is brought to the emergency department by her mother because of a 2-hour history of severe vomiting. Physical examination shows signs of dehydration and yellow discoloration to her skin and sclera. Obstruction of the GI is suspected, and a barium swallow is performed which is shown in the attached image. Which of the following structures is most likely obstructed in this condition?

- A. Pylorus of the stomach
- B. Descending part of duodenum
- C. Horizontal part of duodenum
- D. Duodenum cap(bulb)
- E. Proximal part of jejunum



15. A 35-year-old man comes to the emergency department because of a 2-hour history of epigastric abdominal pain that radiates to the back. His blood pressure is 100/60mmHg, respirations are 21/min and pulse is 100/min. Physical examination shows a diaphoretic patient with abdominal wall guarding and rebound tenderness. Laboratory studies show elevated serum lipase and amylase. Which of the following nerves most likely carries the pain fibers from the affected organ back to the spinal cord?

- A. Lumbar splanchnics
- B. Vagus
- C. Pelvic splanchnics
- D. Lesser thoracic splanchnics
- E. Least thoracic splanchnics
- F. Great thoracic splanchnics
- G. Subcostal
- H. Intercostal

16. A 60-year-old man is brought to the emergency department by his wife because of a 2-hour history of vomiting blood. He is an alcoholic who has been previously treated for hemorrhoids and jaundice. Physical examination shows varicosities on his anterior abdominal wall radiating from his umbilical area. The anastomosis of which of the following veins is most likely responsible for the appearance of his abdominal wall?

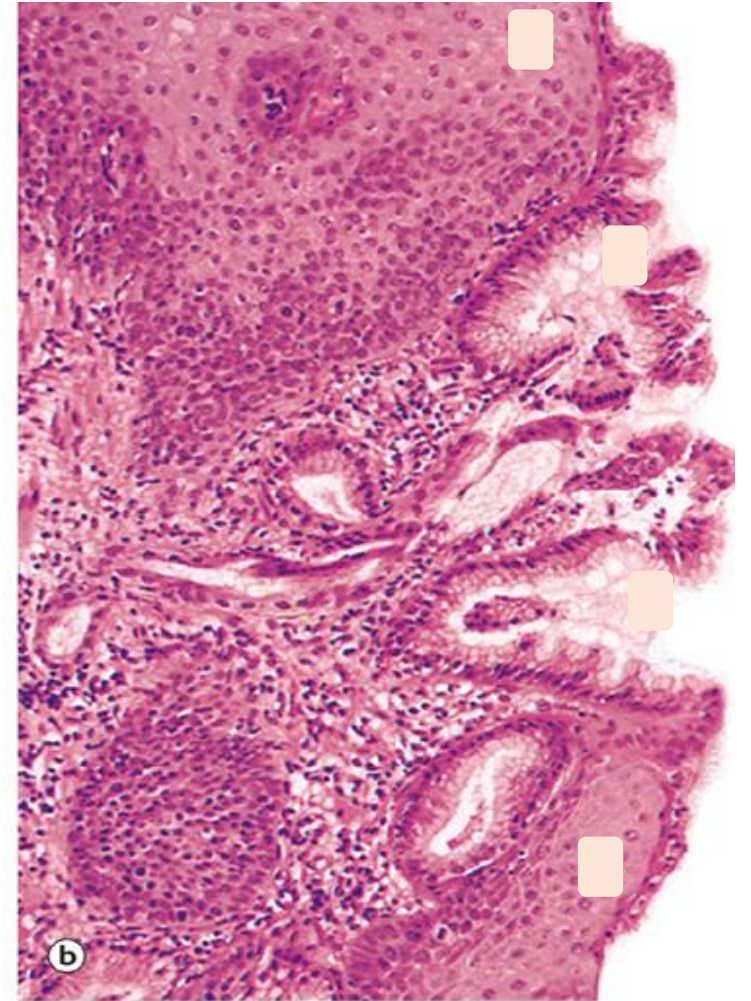
- A. Superior and inferior epigastric
- B. Splenic and left gastric
- C. Portal and inferior vena cava
- D. Esophageal and inferior vena cava
- E. Esophageal and left gastric
- F. Paraumbilical and superficial epigastric

17. A 43-year old woman comes to the physician because of a 2-week history of lower abdominal pain. Physical examination shows a distended, painful abdomen with absent bowel sounds. History shows previous surgeries for gallstones, perforated gastric ulcer and appendicitis. Surgical exploration shows severe intestinal adhesion formation. Which of the following histological layers of the tubular GI tract was most likely damaged during her past surgeries resulting in these adhesions?

- A. Adventitia
- B. Muscularis mucosa
- C. Submucosa
- D. Muscularis externa
- E. Lamina propria
- F. Serosa

18. A 43-year-old man comes to the emergency department because of a 1-week history of burning sensation in his chest and difficulty swallowing. He has a 3-year history of heartburn. Physical examination shows no abnormalities. Endoscopy of the upper GI shows inflammation and a biopsy is collected from the lower end of esophagus which is shown. Which of the following cellular adaptations to injury most likely occurred in this patient's condition?

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- B. Metaplasia
- C. Atrophy
- D. Hyperplasia
- E. Dysplasia



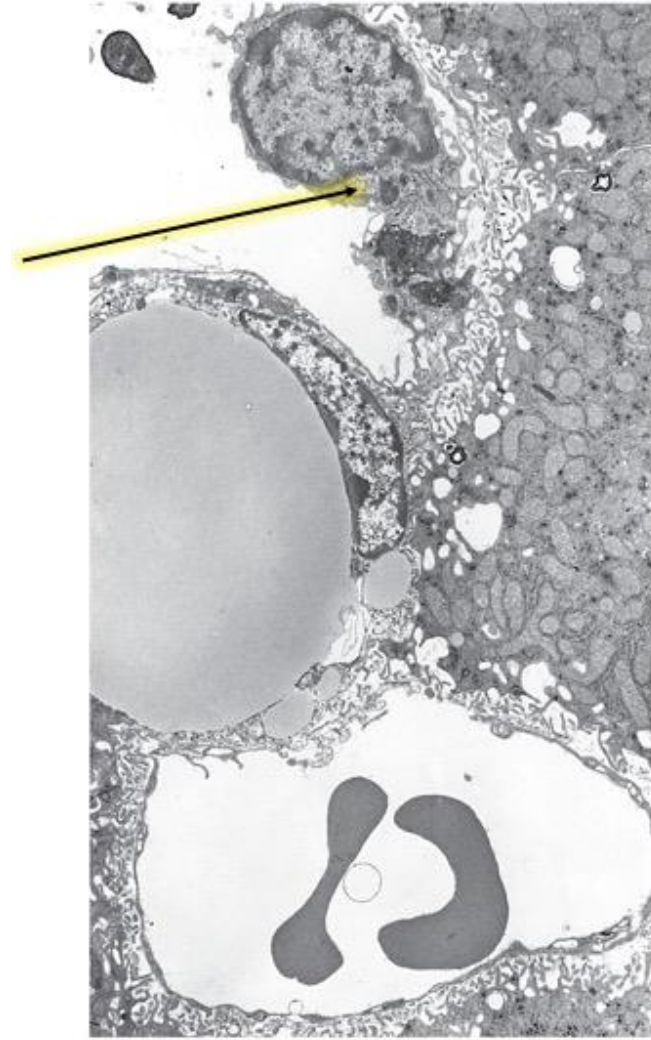
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- E. Contains large number of Paneth cells
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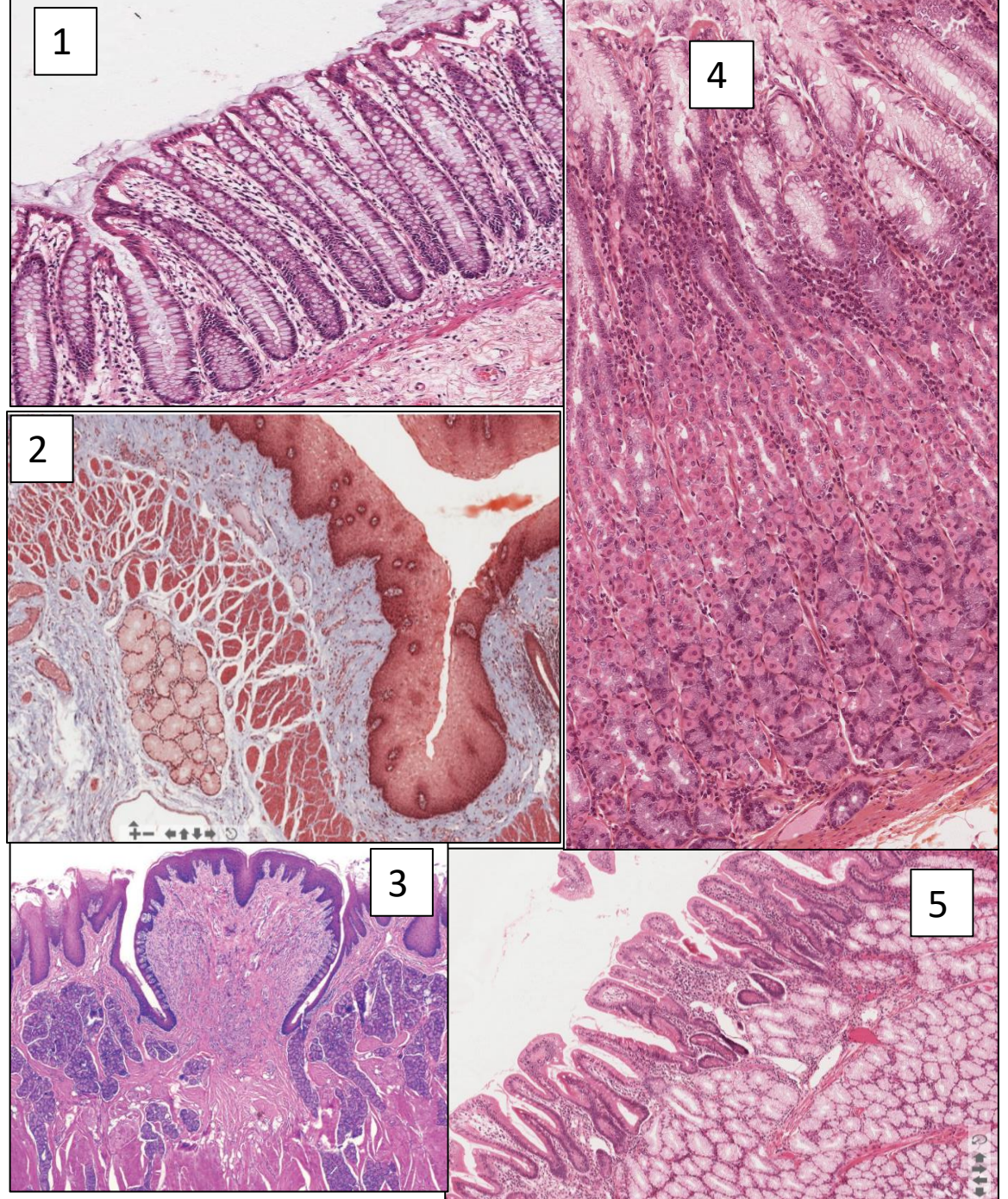
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- B. Hepatic stellate (Ito) cell
- C. Hepatocyte
- D. Cholangiocyte
- E. Kupffer cell



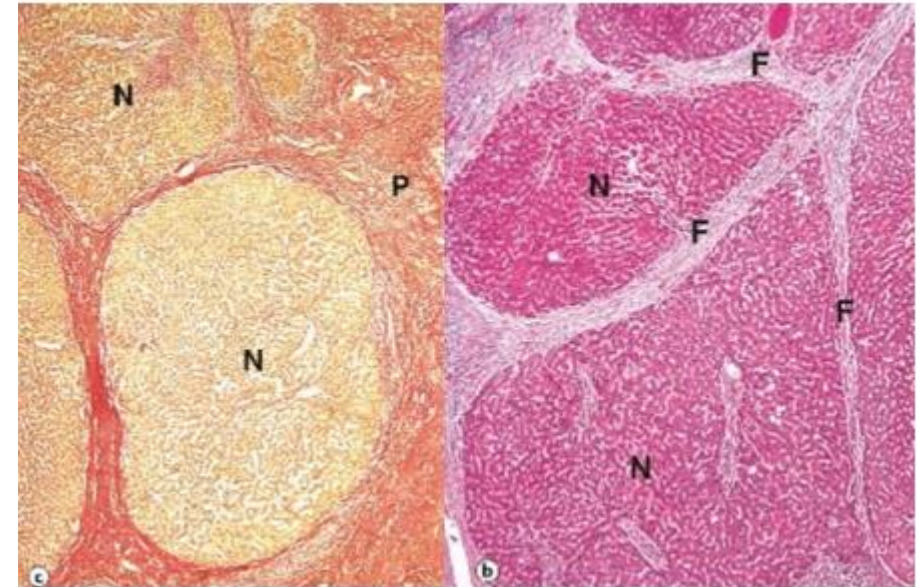
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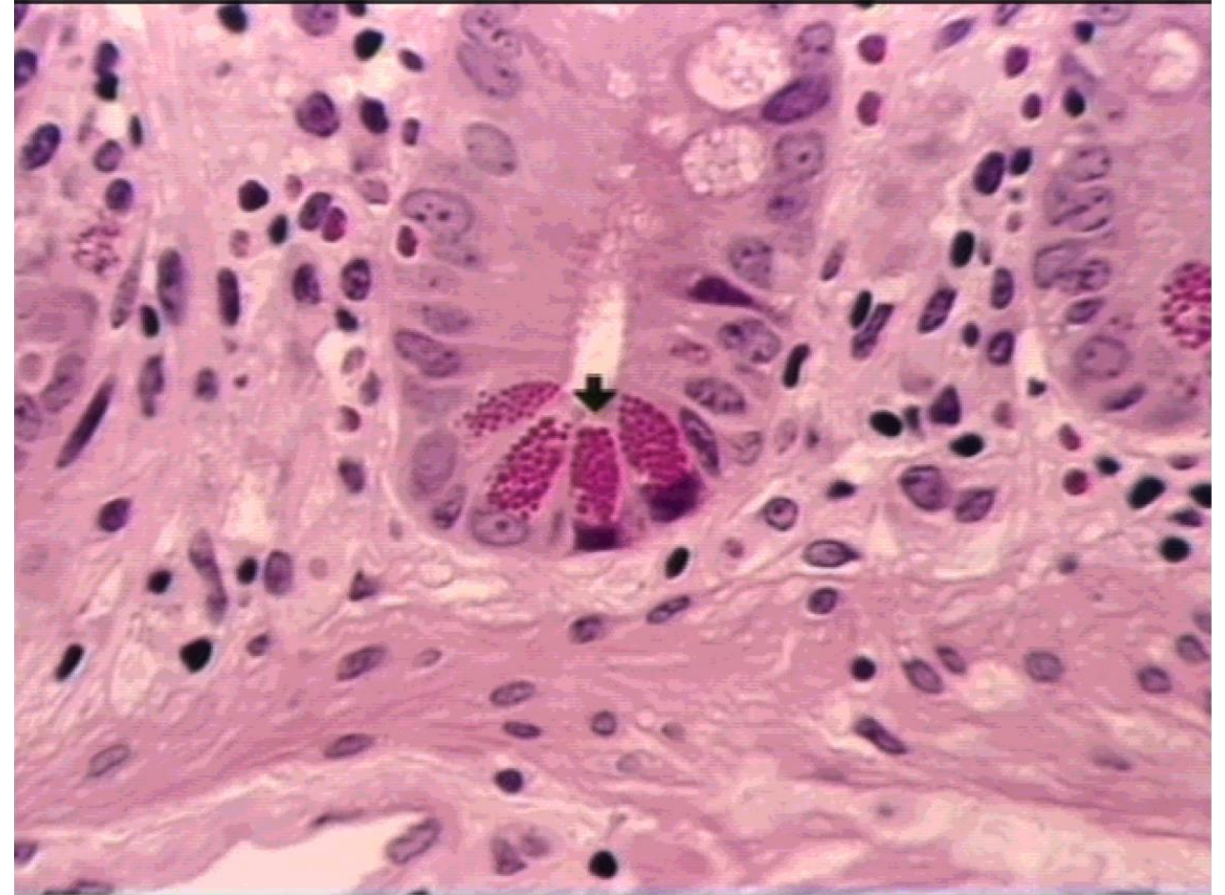
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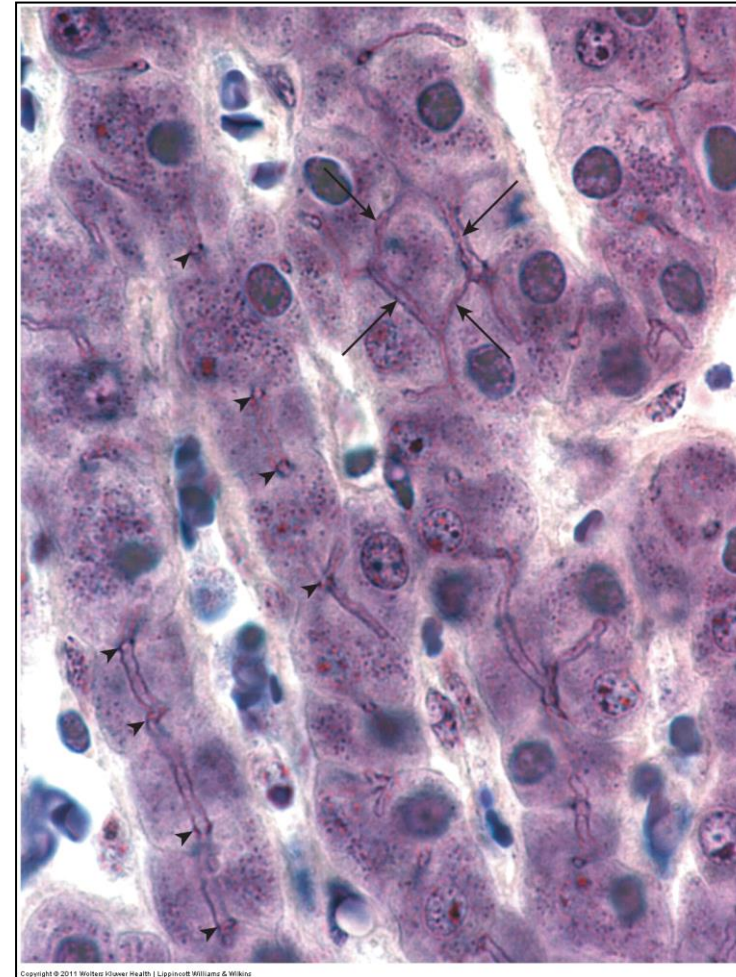
23. A 38-year-old man comes to the physician for an endoscopy. On visualization of small intestine, some areas of the mucosa are found to be inflamed. Biopsy of this region is taken, and the tissue is sent for histopathological examination. The tissue is found to be normal and shows several glands as shown below. Which of the following functions is most likely performed by the cells at the pointer?

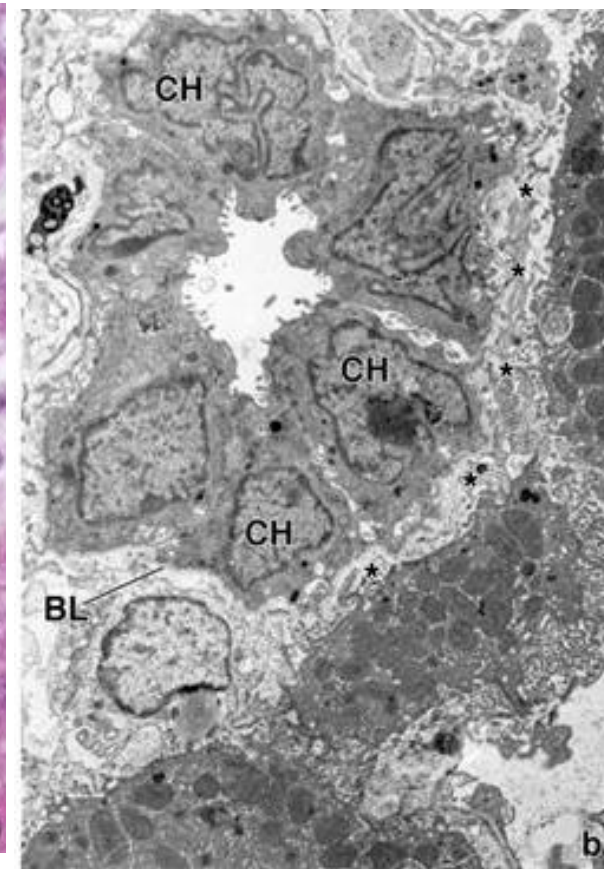
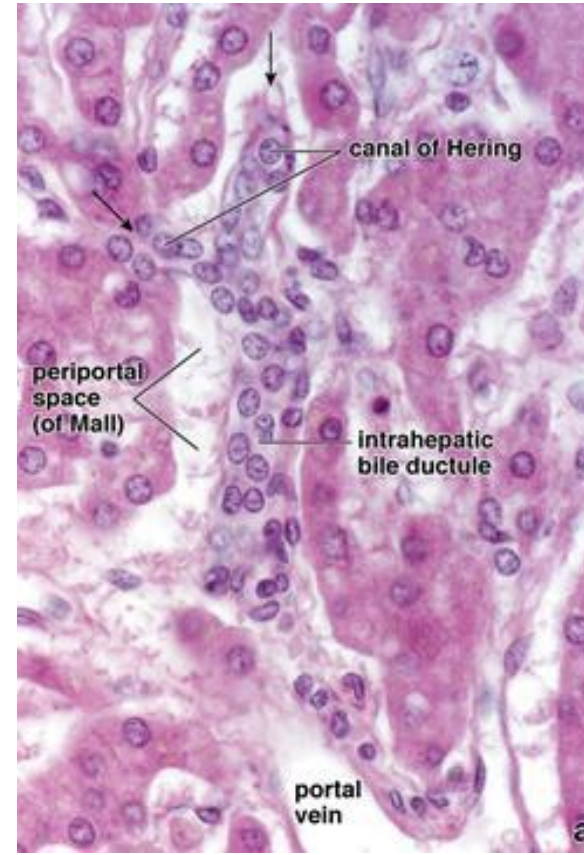
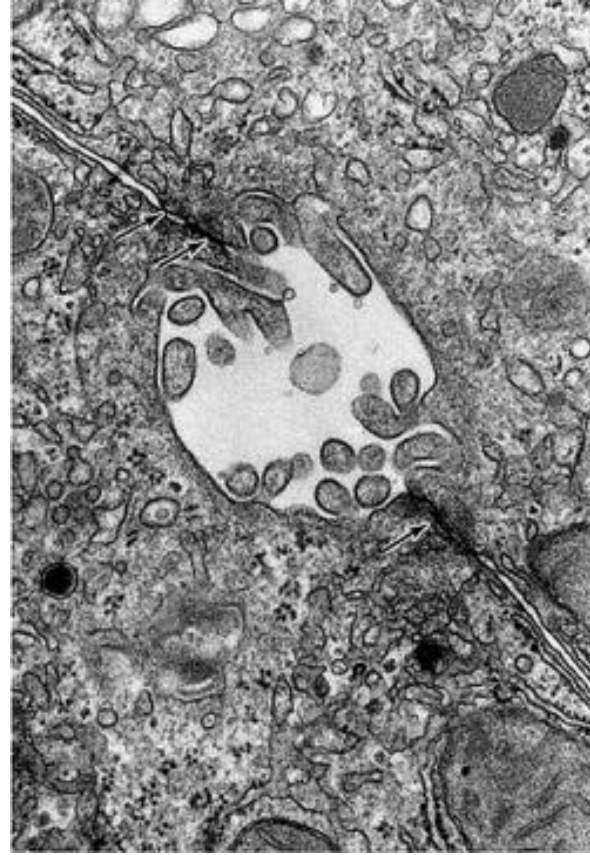
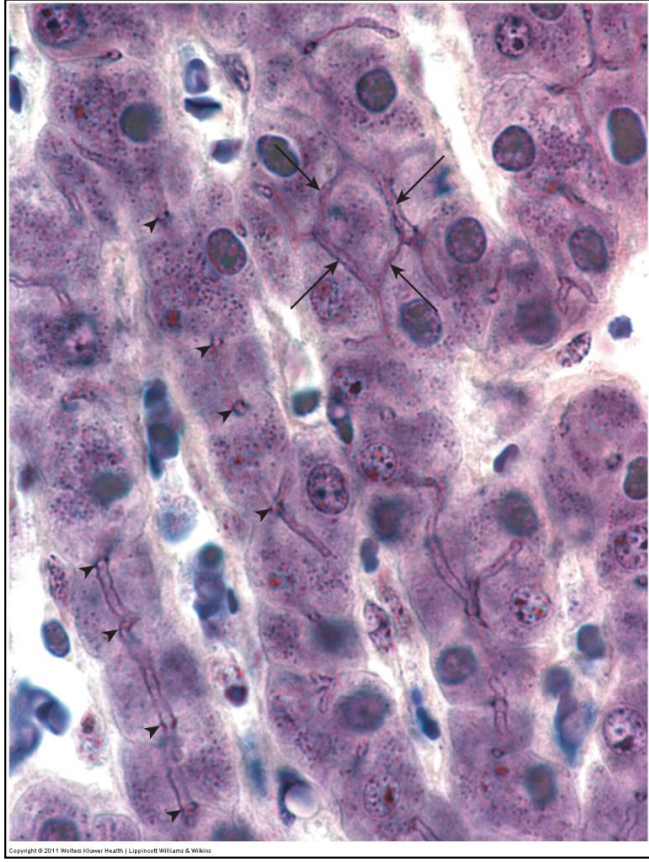
- A. Unicellular mucus secreting cells
- B. Produce digestive enzymes
- C. Produce secretin and motilin
- D. Serve as stem cells of the epithelium
- E. Regulates intestinal normal bacterial flora



24. A 65-year-old man comes to the physician because of a 1-month history of upper abdominal pain. Physical examination shows an enlarged liver and jaundice of mucous membranes. Blood tests show elevated liver enzymes. A liver biopsy shows obstruction of the structures at the arrows & arrowheads. Which of the following cells most likely form these structures?

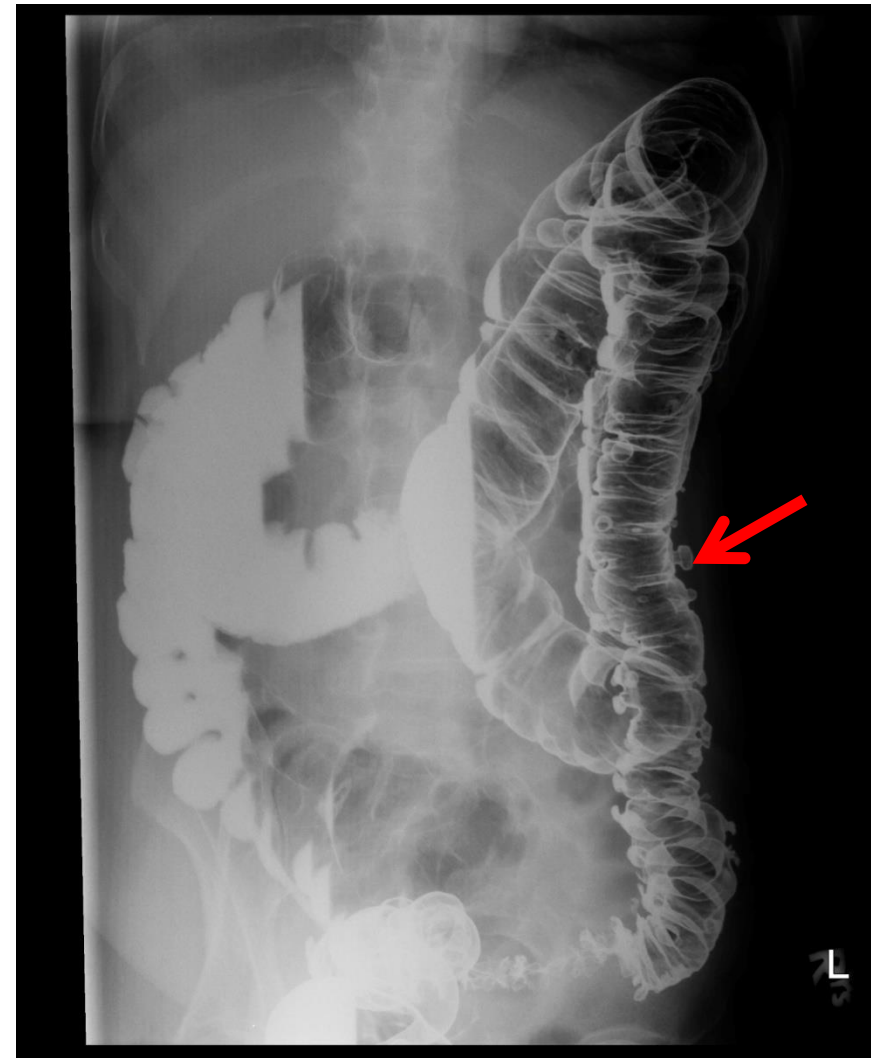
- A. Endothelial cells
- B. Hepatocytes
- C. Hepatic stellate (Ito) cells
- D. Kupffer cells
- E. Cholangiocytes





25. A 60-year-old woman comes to the emergency department because of a 2-day history of fever and severe abdominal pain. Physical examination shows muscle guarding and rebound tenderness on deep palpation. An x-ray and barium enema of her abdomen is shown. Which of the following nerves most likely brings parasympathetic innervation to the part of the GI indicated by the arrow?

- A. Lumbar splanchnics
- B. Pelvic splanchnics
- C. Lesser thoracic splanchnics
- D. Greater thoracic splanchnics
- E. Least thoracic splanchnics
- F. Vagus

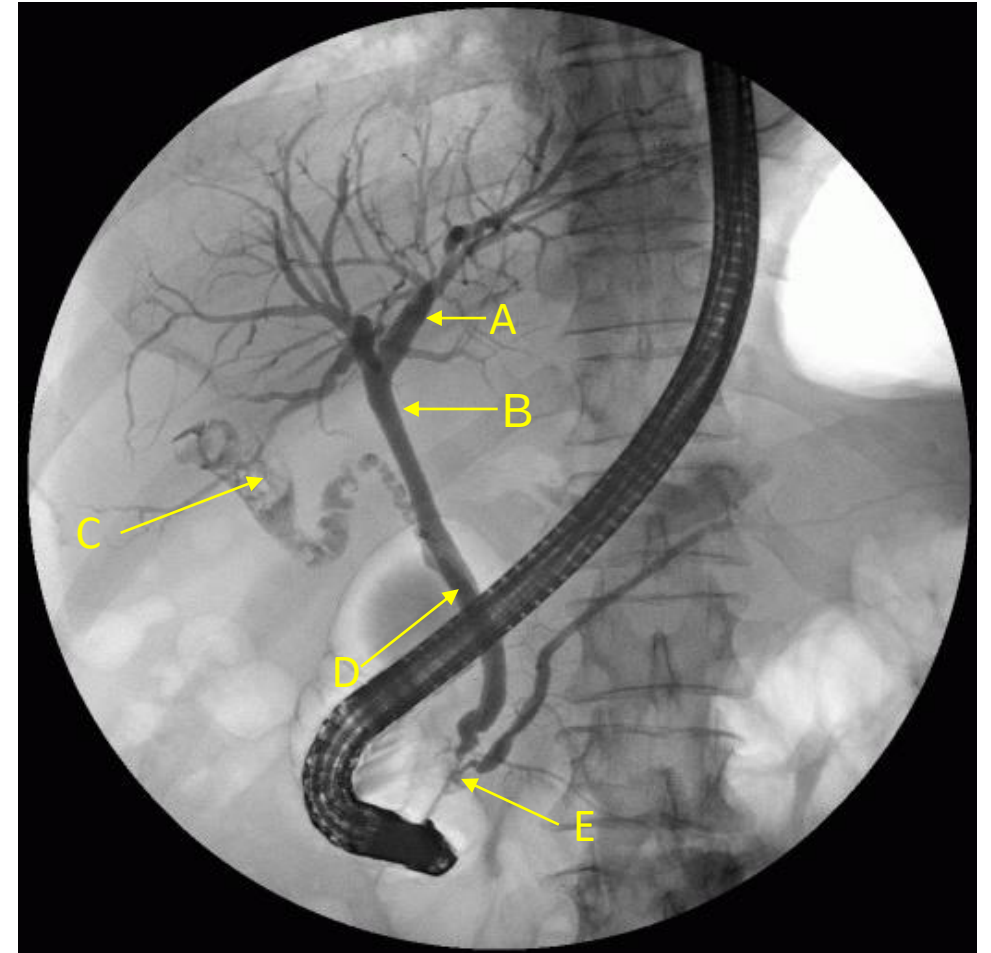


26. A 60-year-old man is brought to the emergency department by his wife because of a 2-hour history of vomiting blood. He is an alcoholic who has been previously treated for hemorrhoids and jaundice. Physical examination shows varicosities on his anterior abdominal wall radiating from his umbilical area. The anastomosis of which of the following veins is most likely responsible for his vomiting?

- A. Superior and inferior epigastric
- B. Splenic and left gastric
- C. Portal and inferior vena cava
- D. Esophageal and inferior vena cava
- E. Esophageal and left gastric
- F. Paraumbilical and superficial epigastric

27. A 40-year-old woman comes to the emergency department because of a 12-hour history of nausea, vomiting, intense itching and pain in the right upper quadrant. Physical examination shows yellow mucous membranes, abdominal tenderness and inspiratory arrest on deep palpation in the right subcostal region. Laboratory studies show elevation of serum bilirubin, markedly increased alkaline phosphatase and serum amylase. Endoscopic retrograde cholangiopancreatography(ERCP) shows an obstruction within the biliary tree. In the image provided, which of the following locations most likely represents the site of obstruction?

- A. A
- B. B
- C. C
- D. D
- E. E



28. A 72-year-old man is brought to the emergency department because of a 1-hour history of severe abdominal pain. His blood pressure is 70/60mmHg, respirations are 22/min and pulse is 100/min. Physical examination shows a pulsatile mass on superficial palpation of the abdomen. CT scan shows a large abdominal aortic aneurysm that includes the origin of the inferior mesenteric artery. During surgical repair of the aneurysm, the inferior mesenteric artery is sacrificed. Which of the following arteries would most likely provide collateral supply to the descending colon?

- A. Left gastroepiploic artery
- B. Sigmoid artery
- C. Splenic artery
- D. Marginal artery
- E. Superior rectal artery

29. A 47-year-old man is admitted to the hospital for surgical repair of a bleeding duodenal ulcer. During this procedure, the gastroduodenal artery is ligated. A branch of which of the following arteries will most likely continue to supply blood to the pancreas in this patient?

- A. Inferior mesenteric
- B. Left gastric
- C. Right gastric
- D. Proper hepatic
- E. Superior mesenteric artery

30. A 62-year-old woman comes to the emergency department because of a 3-hour history of severe chest pain. The pain is a burning sensation in her chest that radiates to her back. Physical examination shows bad breath and inflammation of her gums. Endoscopy shows signs expanding erythema in the lower end of the esophagus. A diagnosis of gastroesophageal reflux disease is made. She undergoes a fundoplication where the fundus of the stomach is completely sutured around the gastroesophageal junction. During the procedure, the posterior vagal trunk is injured while inserting staples in the stomach. Which of the following organs will most likely be affected?

- A. Urinary bladder
- B. Descending colon
- C. Esophagus
- D. Prostate gland
- E. Jejunum

31. A 50-year-old man comes to the physician because of a 1-month history of unintentional weight loss and small amounts of blood in his feces. Physical examination shows an emaciated patient. Colonoscopy shows a cancerous growth in his ascending colon. Which of the following best describes this part of the colon?

- A. It is intraperitoneal
- B. It receives blood supply from the inferior mesenteric artery
- C. It is enclosed within the sigmoid mesocolon
- D. Its sympathetic innervation is from lumbar splanchnics
- E. Its parasympathetic innervation is from the pelvic splanchnics
- F. Pain fibers from this part of the colon travel in the least thoracic splanchnics

32. A 54-year-old man comes to the emergency department because of a 2-hour history of severe bile-stained vomit. Endoscopic examination shows part of the bowel being compressed between the aorta and the superior mesenteric artery. Which part of the GI is most likely being compressed?

- A. Descending part of duodenum
- B. Ileum
- C. Horizontal part of duodenum
- D. Duodenum cap(bulb)
- E. Proximal part of jejunum

Key

- | | |
|-------|-------|
| 1. A | |
| 2. B | 17. F |
| 3. C | 18. B |
| 4. D | 19. E |
| 5. D | 20. E |
| 6. B | 21. D |
| 7. E | 22. F |
| 8. C | 23. E |
| 9. A | 24. B |
| 10. D | 25. B |
| 11. B | 26. E |
| 12. D | 27. E |
| 13. D | 28. D |
| 14. B | 29. E |
| 15. F | 30. E |
| 16. F | 31. F |
| | 32. C |